YOUTUBE-DNN-RECSYS — SYSTEM ARCHITECTURE

Covington two-tower recall · source: docs/architecture.drawio · github.com/chenxingqiang/YouTube-DNN-RecSys

DATA PROCESSING

data2tfrecords.py
src/python/data/data2tfrecords.py
random + TFRecords train/eval

read_tfrecords.py
src/python/data/read_tfrecords.py

MakeDataOne.scala
src/scala/data/MakeDataOne.scala
Spark distributed generate

MakeDataTwo.scala
src/scala/data/MakeDataTwo.scala
HDFS TFRecords

STORAGE

TFRecords

data/tfrecords/
train + eval TFRecords

data/checkpoints/
TF ckpt + SavedModel

MODEL src/python/models/dnn.py (TensorFlow Estimator)

Input
visit_items_index (5)
continuous_features (16)

Embedding
item_table-54
average pooling on history

SavedModel

MLP tower
dense layers → user vector u

Softmax
500-way next-item
cross entropy · Top-K

PREDICTION

load_dnn_model.py
src/python/models/load_dnn_model.py
SavedModel online / batch

PredictUserVector.scala
src/scala/prediction/PredictUserVector.scala

ItemEmbeddingPredictor.scala
src/scala/prediction/ItemEmbeddingPredictor.scala

ItemEmbedding.scala
src/scala/models/ItemEmbedding.scala

UTILITIES & CONFIG

tensor_board.py
src/python/utils/tensor_board.py

feature_column.py
src/python/reference/feature_column.py

self_defined_network_layer.py
src/python/reference/self_defined_network_layer.p

FeatureBuilder.scala
src/scala/models/FeatureBuilder.scala

BaseSparkLocal/Online
src/scala/core/*.scala

vars.prod/sit.properties
src/scala/config/

Python/TF path and Scala/Spark path share TFRecords + the same two-tower (history pool → user vec · item table)